Type MS · Never at rest, and the least speed

Answer 1 of 4 · Question 7 part d.ii · 3 marks

2023 JAN · Paper 01 (regional R) · extract 1 of 3 · find part d.ii in this mark scheme extract

7(a)	$v = 1^2 - 10\times1 + 28 = 19$ [m/s]	B1 [1]
(b)	$s = \int\left(t^2 - 10t + 28\right)\,\mathrm{d}t = \frac{t^3}{3} - \frac{10t^2}{2} + 28t\,(+c)$ $24 = \frac{3^3}{3} - \frac{10\times3^2}{2} + 28\times3 + c \Rightarrow c = -24 \Rightarrow \left[s = \frac{t^3}{3} - \frac{10t^2}{2} + 28t - 24\right]$ $t = 5, \quad s = \frac{5^3}{3} - \frac{10\times5^2}{2} + 28\times5 - 24 = \frac{98}{3}$ [m]	M1 M1A1 M1A1 [5]
(c)	$\frac{\mathrm{d}v}{\mathrm{d}t} = 2t - 10$ when $t = 9$, acceleration = 8 [m/s ^{2}]	M1 A1 [2]
(d)	(i) $v = (t-5)^2 + 3$ Irrespective of the value of t $v \geqslant 3$ so the particle never comes to rest. **ALT** $b^2 - 4ac < 0 \Rightarrow (-10)^2 - 4\times1\times28 = -12$ No real solutions so the particle never comes to rest. (ii) Least value of v is 3 [m/s]	M1 A1 [M1 A1] B1 [3]
		Total 11 marks

Question	Notes	Marks
7(a)	$v = t^2 - 10t + 28$	
	$v = 1^2 - 10\times1 + 28 = 19$ [m/s]	B1 [1]
(b)	For an attempt to integrate the given expression for v [See general guidance for the definition of an attempt] $s = \int\left(t^2 - 10t + 28\right)\,\mathrm{d}t = \frac{t^3}{3} - \frac{10t^2}{2} + 28t\,(+c)$	M1
	For finding the value of c. They cannot score this mark without $+c$ $24 = \frac{3^3}{3} - \frac{10\times3^2}{2} + 28\times3 + c \Rightarrow c = \ldots$	M1
	For the correct expression for s. This need not be explicitly stated. $s = \frac{t^3}{3} - \frac{10t^2}{2} + 28t - 24$	A1

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Answer 1 of 4 · Question 7 part d.ii · 3 marks

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	Award for the correct value of c seen -24 [m]	
	For using their integrated expression for s to find a value of the displacement when $t = 5$ $s = \frac{5^3}{3} - \frac{10\times5^2}{2} + 28\times5 - 24 = ...$	M1
	For the correct value of $s = \frac{98}{3}$ [m] Accept $s = 32.7$ [or better]	A1 [5]
(c)	For an attempt to differentiate the given v and substituting $t = 9$ into their differentiated expression. $\frac{dv}{dt} = 2t - 10 \Rightarrow \frac{dv}{dt} = 2\times9 - 10 = ...$	M1
	For acceleration = 8 [m/s ^{2}]	A1 [2]
(d)(i)	**Method A** Completes the square to give $v = (t-5)^2 + 3$	M1
	Concludes that at the minimum velocity [3 m/s] $t = 5$ so P never comes to rest	A1
	Method B Finds the value of the discriminant $b^2 - 4ac < 0 \Rightarrow (-10)^2 - 4\times1\times28 = -12$	M1
	Concludes that as there are no real solutions, **so *P* does not come to rest.**	A1
	Method C Solves the 3TQ to give the following two [non-real] values of t: $5+\sqrt{3}\text{i}$ and $5-\sqrt{3}\text{i}$ or $\left(t-\left[5+\sqrt{3}\text{i}\right]\right)\left(t-\left[5+\sqrt{3}\text{i}\right]\right)=0$	M1
	Concludes that as there are no real solutions, **so *P* does not come to rest.**	A1
	Method D Uses their result from (c) $\frac{dv}{dt} = 2t - 10 = 0 \Rightarrow t = 5$	M1
	Concludes that at the minimum velocity [3 m/s] $t = 5$ so P never comes to rest	A1
(ii)	For the correct value of $v = 3$ [m/s}	B1 [3]
		Total 11 marks

Type MS · Never at rest, and the least speed

Answer 1 of 4 · Question 7 part d.ii · 3 marks

2023 JAN · Paper 01 (regional R) · extract 3 of 3 · find part d.ii in this mark scheme extract

Question	Scheme	Marks

Type MS · Never at rest, and the least speed

Answer 2 of 4 · Question 1 part a · 2 marks

2025 JUN · Paper 01 · extract 1 of 2 · find part a in this mark scheme extract

1 (a)	$b^2-4ac=(-9)^2-4\times3\times7=-3$ $-3<0$ so no (real) solutions for v	M1 A1 [2]
(b)	$\left\{\frac{dv}{dt}\right\}=6t-9\Rightarrow a=6\times4-9=15\left(m/s^2\right)$	M1A1 [2]
(c)	$s=\int_0^5(3t^2-9t+7)\,dt=\left[t^3-\frac{9}{2}t^2+7t\right]_0^5$ $=5^3-\frac{9}{2}\times5^2+7\times5=\frac{95}{2}$ (m) oe	M1 M1A1 [3]
		Total 7 marks

Part	Mark	Additional Guidance
(a)	M1	Applies the discriminant, or any other valid method. e.g. Attempts to solve the 3TQ by quadratic formula, Completing square, achieves $3\left(t-\frac{3}{2}\right)^2+...$ Attempts to solve $v = 0$, finds complex roots
	A1	Verifies that the equation for v has no (real) solutions By completing square, it must be correct, $3\left(t-\frac{3}{2}\right)^2+\frac{1}{4}$, states the graph is above x axis, or > 0 Finds correct complex roots $\frac{9\pm\sqrt{3}i}{6}$, states imaginary numbers/roots **Note**: Just stating "error" is not enough
(b)	M1	Attempts to differentiate the expression for v. Power of t to decrease in at least one term and increase in none.
	A1	Substitutes $t = 4$ and obtains correct acceleration, with or without unit.
(c)	M1	Attempt to integrate the expression for v. Power of t to increase in at least one term and decrease in none. Ignore limits if shown.
	M1	Clear substitution of the **given** limits into their **integrated** expression for distance, no need to see substitution of zero, can be implied by the correct distance found from a **correct integrated** expression
	A1	$\frac{95}{2}$ (m) oe e.g. 47.5, ignore unit

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Answer 2 of 4 · Question 1 part a · 2 marks

2025 JUN · Paper 01 · extract 2 of 2 · find part a in this mark scheme extract

Question number	Scheme	Marks

Type MS · Never at rest, and the least speed

Answer 3 of 4 · Question 4 part a · 3 marks

2025 NOV · Paper 02 · extract 1 of 3 · find part a in this mark scheme extract

Part	Mark	Notes
(a)	B1	For $\frac{4x-3}{2x+1}$ or $\frac{2(2x+1)}{2x+1}-\frac{5}{2x+1}$ or $\frac{4x+2-5}{2x+1}$ **Both** correct asymptote equations seen will imply this mark.
(i)	B1	For $x=-\frac{1}{2}$ Independent of method accuracy mark, **so ignore any incorrect attempt to form a single fraction**
(ii)	B1	For $y=2$ Independent of method accuracy mark, **so ignore any incorrect attempt to form a single fraction**
(b)	B1ft	For the coordinates of the y-intercept, follow through their a or b and c from any expression of the form $\frac{4x+a}{2x+1}$ $a\neq 0$ or $\frac{4x+b+c}{2x+1}$ $b,c\neq 0$ Condone eg just $y=-3$ or $y=a$ without $x=0$ shown.
	B1ft	For the coordinates of the x-intercept, follow through their a as above. ie ft $(x=)-\frac{a}{4}$ Condone $x=\frac{3}{4}$ or $x=-\frac{a}{4}$ without $y=0$ shown.
Note on labelling: Answers presented in the correct order without labelling – we would award all marks. If labelling not present, we can't assume the candidate's thinking.		
(c)	B1	For a negative reciprocal curve drawn anywhere in the grid – there **must** be two branches present, they **must not** cross any asymptotes drawn or implied or cross each other and **must not obviously** 'bend back' on themselves. Mark intention.
	B1ft	Two clearly marked asymptotes, ft their asymptotes listed from part (a). Labelled as equations or clearly passing through correctly labelled numbers on the axes. Must be one section of negative reciprocal curve, tending towards these asymptotes. This branch must not obviously cross or bend back from any asymptotes drawn or implied. Mark intention. Ignore any other branch/curve/line drawn if there is one branch fulfilling these conditions.
	B1ft	Single curve (or even line) passing through the correct intercepts. Ft their coordinates from part (b) marked clearly on the graph as coordinates or crossing points on the axes. Ignore any other branch/curve/line present.
	The answers to (a) and (b) cannot be retrieved from (c)	

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Answer 3 of 4 · Question 4 part a · 3 marks

2025 NOV · Paper 02 · extract 2 of 3 · find part a in this mark scheme extract

Question	Scheme	Marks
4(a)	$\left(\frac{dv}{dt}=\right)8t-6=0\Rightarrow t=\left(\frac{6}{8}\right)$ oe	M1
	$v=4\times\left("\frac{6}{8}"\right)^2-6\times\left("\frac{6}{8}"\right)+5=\frac{11}{4}[\text{m/s}]$ oe	dM1A1 [3]
ALT	$4\left(t-\frac{3}{4}\right)^2+\frac{11}{4}$ oe	M1 M1
	$\frac{11}{4}[\text{m/s}]$ oe	A1 [3]
(b)	$18=8T-6\Rightarrow T=3$	M1A1 [2]
(c)	$\left[XY=\int_0^3\left(4t^2-6t+5\right)dt=\right]$	
	$\left\|\left[\frac{4t^3}{3}-\frac{6t^2}{2}+5t\right]_0^3\right\|$	M1A1
	$XY=\left(\frac{4\times3^3}{3}-\frac{6\times3^2}{2}+5\times3\right)-(0)=24\,(\text{m})$	M1A1 [4]
		Total 9 marks

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Answer 3 of 4 · Question 4 part a · 3 marks

2025 NOV · Paper 02 · extract 3 of 3 · find part a in this mark scheme extract

Question	Scheme	Marks

Type MS · Never at rest, and the least speed

Answer 4 of 4 · Question 7 part d.i · 0 marks

2023 JAN · Paper 01 (regional R) · extract 1 of 3 · find part d.i in this mark scheme extract

7(a)	$v = 1^2 - 10\times1 + 28 = 19$ [m/s]	B1 [1]
(b)	$s = \int\left(t^2 - 10t + 28\right)\,\mathrm{d}t = \frac{t^3}{3} - \frac{10t^2}{2} + 28t\,(+c)$ $24 = \frac{3^3}{3} - \frac{10\times3^2}{2} + 28\times3 + c \Rightarrow c = -24 \Rightarrow \left[s = \frac{t^3}{3} - \frac{10t^2}{2} + 28t - 24\right]$ $t = 5, \quad s = \frac{5^3}{3} - \frac{10\times5^2}{2} + 28\times5 - 24 = \frac{98}{3}$ [m]	M1 M1A1 M1A1 [5]
(c)	$\frac{\mathrm{d}v}{\mathrm{d}t} = 2t - 10$ when $t = 9$, acceleration = 8 [m/s ^{2}]	M1 A1 [2]
(d)	(i) $v = (t-5)^2 + 3$ Irrespective of the value of t $v \geqslant 3$ so the particle never comes to rest. **ALT** $b^2 - 4ac < 0 \Rightarrow (-10)^2 - 4\times1\times28 = -12$ No real solutions so the particle never comes to rest. (ii) Least value of v is 3 [m/s]	M1 A1 [M1 A1] B1 [3]
		Total 11 marks

Question	Notes	Marks
7(a)	$v = t^2 - 10t + 28$	
	$v = 1^2 - 10\times1 + 28 = 19$ [m/s]	B1 [1]
(b)	For an attempt to integrate the given expression for v [See general guidance for the definition of an attempt] $s = \int\left(t^2 - 10t + 28\right)\,\mathrm{d}t = \frac{t^3}{3} - \frac{10t^2}{2} + 28t\,(+c)$	M1
	For finding the value of c. They cannot score this mark without $+c$ $24 = \frac{3^3}{3} - \frac{10\times3^2}{2} + 28\times3 + c \Rightarrow c = \ldots$	M1
	For the correct expression for s. This need not be explicitly stated. $s = \frac{t^3}{3} - \frac{10t^2}{2} + 28t - 24$	A1

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Answer 4 of 4 · Question 7 part d.i · 0 marks

2023 JAN · Paper 01 (regional R) · extract 2 of 3 · find part d.i in this mark scheme extract

	Award for the correct value of c seen -24 [m]	
	For using their integrated expression for s to find a value of the displacement when $t = 5$ $s = \frac{5^3}{3} - \frac{10\times5^2}{2} + 28\times5 - 24 = ...$	M1
	For the correct value of $s = \frac{98}{3}$ [m] Accept $s = 32.7$ [or better]	A1 [5]
(c)	For an attempt to differentiate the given v and substituting $t = 9$ into their differentiated expression. $\frac{dv}{dt} = 2t - 10 \Rightarrow \frac{dv}{dt} = 2\times9 - 10 = ...$	M1
	For acceleration = 8 [m/s ^{2}]	A1 [2]
(d)(i)	**Method A** Completes the square to give $v = (t-5)^2 + 3$	M1
	Concludes that at the minimum velocity [3 m/s] $t = 5$ so P never comes to rest	A1
	Method B Finds the value of the discriminant $b^2 - 4ac < 0 \Rightarrow (-10)^2 - 4\times1\times28 = -12$	M1
	Concludes that as there are no real solutions, **so *P* does not come to rest.**	A1
	Method C Solves the 3TQ to give the following two [non-real] values of t: $5+\sqrt{3}\mathrm{i}$ and $5-\sqrt{3}\mathrm{i}$ or $\left(t-\left[5+\sqrt{3}\mathrm{i}\right]\right)\left(t-\left[5+\sqrt{3}\mathrm{i}\right]\right)=0$	M1
	Concludes that as there are no real solutions, **so *P* does not come to rest.**	A1
	Method D Uses their result from (c) $\frac{dv}{dt} = 2t - 10 = 0 \Rightarrow t = 5$	M1
	Concludes that at the minimum velocity [3 m/s] $t = 5$ so P never comes to rest	A1
(ii)	For the correct value of $v = 3$ [m/s}	B1 [3]
		Total 11 marks

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Answer 4 of 4 · Question 7 part d.i · 0 marks

2023 JAN · Paper 01 (regional R) · extract 3 of 3 · find part d.i in this mark scheme extract

Question	Scheme	Marks